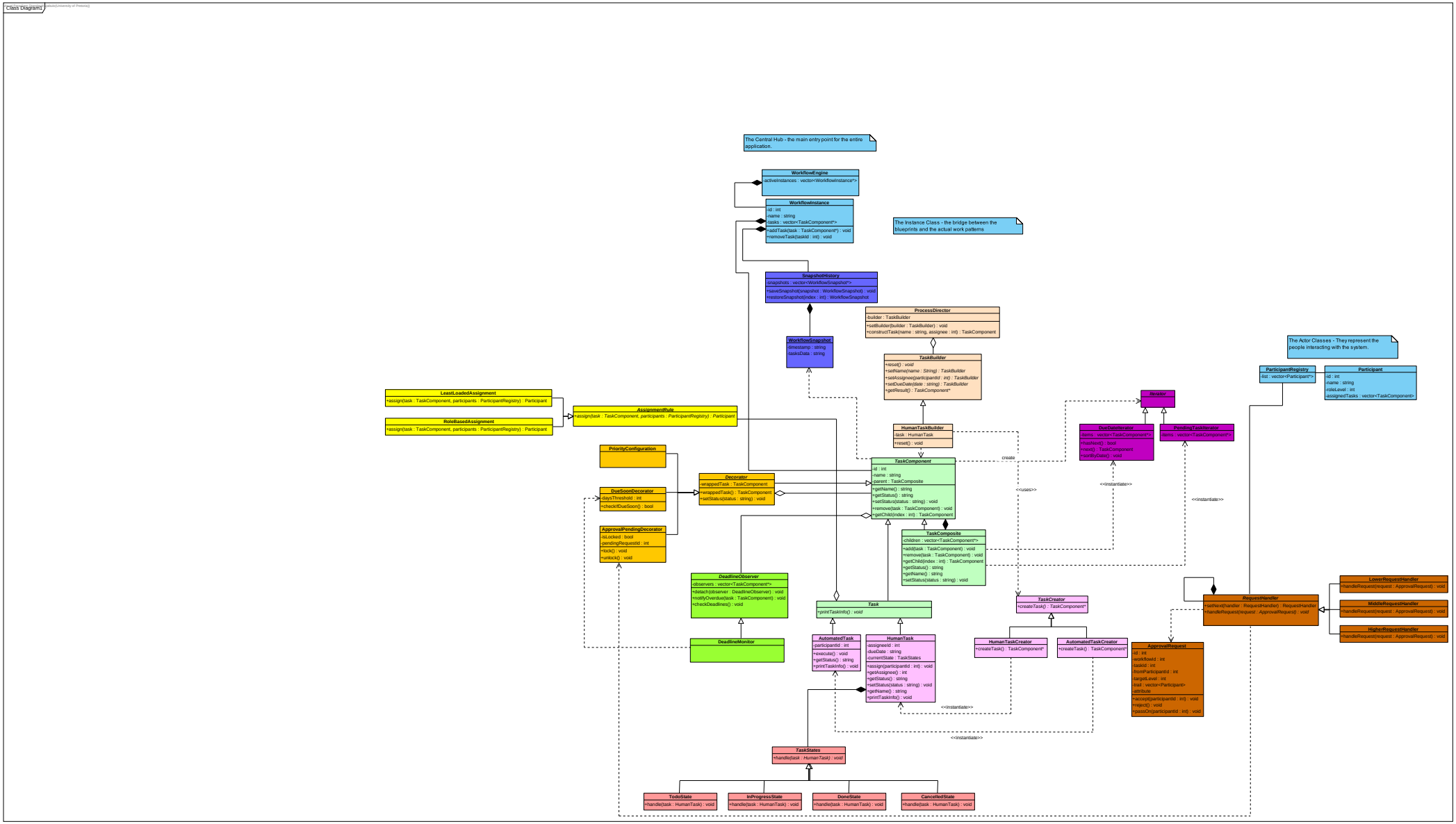
Class Diagram1
The Central Hub - the main entry point for the entire application.
WorkflowEngine
-activeInstances : vector<WorkflowInstance*>
WorkflowInstance
-id : int
-name : string
-tasks : vector<TaskComponent*>
+addTask(task : TaskComponent*) : void
+removeTask(taskId : int) : void
The Instance Class - the bridge between the blueprints and the actual work patterns
SnapshotHistory
-snapshots : vector<WorkflowSnapshot*>
+saveSnapshot(snapshot : WorkflowSnapshot) : void
+restoreSnapshot(index : int) : WorkflowSnapshot
WorkflowSnapshot
-timestamp : string
-tasksData : string
ProcessDirector
-builder : TaskBuilder
+setBuilder(builder : TaskBuilder) : void
+constructTask(name : string, assignee : int) : TaskComponent
TaskBuilder
+reset() : void
+setName(name : String) : TaskBuilder
+setAssignee(participantId : int) : TaskBuilder
+setDueDate(date : string) : TaskBuilder
+getResult() : TaskComponent*
HumanTaskBuilder
-task : HumanTask
+reset() : void
The Actor Classes - They represent the people interacting with the system.
ParticipantRegistry
-list : vector<Participant*>
Participant
-id : int
-name : string
+roleLevel : int
-assignedTasks : vector<TaskComponent>
Iterator
DueDateIterator
-items : vector<TaskComponent*>
+hasNext() : bool
+next() : TaskComponent
+sortByDate() : void
PendingTaskIterator
-items : vector<TaskComponent*>
LeastLoadedAssignment
+assign(task : TaskComponent, participants : ParticipantRegistry) : Participant
RoleBasedAssignment
+assign(task : TaskComponent, participants : ParticipantRegistry) : Participant
AssignmentRule
+assign(task : TaskComponent, participants : ParticipantRegistry) : Participant
PriorityConfiguration
Decorator
-wrappedTask : TaskComponent
+wrappedTask() : TaskComponent
-setStatus(status : string) : void
DueSoonDecorator
-daysThreshold : int
+checkIfDueSoon() : bool
ApprovalPendingDecorator
-isLocked : bool
-pendingRequestId : int
+lock() : void
+unlock() : void
TaskComponent
-id : int
-name : string
-parent : TaskComposite
+getName() : string
+getStatus() : string
+setStatus(status : string) : void
+remove(task : TaskComponent) : void
+getChild(index : int) : TaskComponent
create
<<uses>>
<<instantiate>>
<<instantiate>>
TaskComposite
-children : vector<TaskComponent*>
+add(task : TaskComponent) : void
+remove(task : TaskComponent) : void
+getChild(index : int) : TaskComponent
+getStatus() : string
+getName() : string
+setStatus(status : string) : void
DeadlineObserver
-observers : vector<TaskComponent*>
+detach(observer : DeadlineObserver) : void
+notifyOverdue(task : TaskComponent) : void
+checkDeadlines() : void
DeadlineMonitor
Task
+printTaskInfo() : void
TaskCreator
+createTask() : TaskComponent*
HumanTaskCreator
+createTask() : TaskComponent*
AutomatedTaskCreator
+createTask() : TaskComponent*
AutomatedTask
-participantId : int
+execute() : void
+getStatus() : string
+printTaskInfo() : void
HumanTask
-assigneeId : int
-dueDate : string
-currentState : TaskStates
+assign(participantId : int) : void
+getAssignee() : int
+getStatus() : string
+setStatus(status : string) : void
+getName() : string
+printTaskInfo() : void
<<instantiate>>
<<instantiate>>
ApprovalRequest
-id : int
-workflowId : int
-taskId : int
-fromParticipantId : int
-targetLevel : int
-trail : vector<Participant>
-attribute
+accept(participantId : int) : void
+reject() : void
+passOn(participantId : int) : void
RequestHandler
+setNext(handler : RequestHandler) : RequestHandler
+handleRequest(request : ApprovalRequest) : void
LowerRequestHandler
+handleRequest(request : ApprovalRequest) : void
MiddleRequestHandler
+handleRequest(request : ApprovalRequest) : void
HigherRequestHandler
+handleRequest(request : ApprovalRequest) : void
TaskStates
+handle(task : HumanTask) : void
TodoState
+handle(task : HumanTask) : void
InProgressState
+handle(task : HumanTask) : void
DoneState
+handle(task : HumanTask) : void
CancelledState
+handle(task : HumanTask) : void